P–S–602

DECEMBER 13, 1955.

SUPERSEDING
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Revision—1
September 20, 1954

FEDERAL SPECIFICATION

SOAP; MEDIUM TITER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers soap intended primarily for use in cleaning wool fabrics, other fabrics, finished surfaces, leather, and other articles, where comparatively moderate-temperature washing solutions are indicated.

1.2 Classification. — Soap covered by this specification shall be of one type and the following forms:

Form 1.—Chip or flake
Form 2.—Granular or powdered

2. APPLICABLE SPECIFICATIONS, STANDARD, AND OTHER PUBLICATIONS

2.1 Specifications and standard.—The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

P–S–536—Soap and Soap-Products (Including Synthetic Detergents); Methods of Sampling and Testing.

UU–S–48—Sacks, Paper, Shipping.

PPP–D–723—Drums, Fiber.

PPP–P–31—Preservation, Packaging, Packing, and Marking of Material; and Shipping Instructions; (Domestic Shipment and Storage).

PPP–P–32—Preservation, Packaging, Packing, and Marking of Material; and Shipping Instructions; (Domestic Shipments for Immediate Use).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publication.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

American Oil Chemists' Society

Tentative Method Da 30–53 of the Official and Tentative Methods of the American Oil Chemists' Society.

(Copies of Tentative Method Da 30-53 may be obtained from the American Oil Chemists' Society, 35 E. Wacker Drive, Chicago 1, Ill.)

3. REQUIREMENTS

3.1 Composition.—The composition of the soap shall conform to the requirements shown in table I when tested as specified in 4.4.1.

TABLE I.—*Composition*

	Minimum	Maximum
Moisture and matter volatile at 105° C. ....percent	...	7.0
Sum of free alkali or free acid, total matter insoluble in alcohol, and sodium chloride ....percent	...	6.0
Sodium silicate .....percent	0.25	4.0
Free alkali, calculated as sodium hydroxide (NaOH) ....percent	...	0.2
Free acid, calculated as oleic acid ....percent	...	0.2
Matter insoluble in water percent	...	1.5
Rosin	...	none
Sugar	...	none
Copper, parts per million ..	...	10.0
Anhydrous soap ....percent	88.0	...
Titer of mixed fatty acids prepared from the soap ..	18°C.	33°C.
Iodine number (Wijs) of the mixed fatty acids prepared from the soap	75.0	135.0
Residue retained on a No. 12 sieve (Form 2 only) percent	...	1.5
Organic oxidation inhibitor[1] percent	0.05	...

[1] Acceptable inhibitors of the following types shall be used unless otherwise specified: diphenylamine; orthodiphenyl biguanidine; paratertiary amyl phenol; paratertiary butyl phenol. Contractor shall indicate which inhibitor will be used, and shall not change inhibitor during production unless approved by the contracting officer (see 3.4).

3.1.1 *Computation.*—The percentage of moisture and volatile matter shall be computed, and reported by the testing laboratory, on the soap as received. The percentages of all other constituents shall be calculated and reported on an assumed moisture and volatile matter content of 7 percent. (For basis of purchase, see 6.4.)

3.2 Color.—The soap shall have a uniform light color. When specified, the color of the soap furnished under this specification shall conform to that of the accepted bid sample (see 6.2).

3.3 Odor.—The odor shall not be objectionable in the soap as received or in a solution of the soap in water at 120° F. The soap shall not leave an objectionable odor on objects after washing with a water solution of the soap and rinsing thoroughly with hot water. When specified, the odor of the soap, under the above conditions, shall conform to that of the accepted bid sample. (See 6.2.)

3.4 Certificate of compliance.—A certificate of compliance stating that the organic oxidation inhibitor content complies with the requirements of table I shall be furnished the procuring activity.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of not more than 1,000 pounds of soap packaged from the same plant batch and offered for delivery at one time.

4.1.2 *Sampling for tests (at a Government laboratory).*—From each inspection lot (see 4.1.1), the Government inspector shall take two containers at random. From each of the two containers, 1-pound specimens shall be taken and placed in separate clean, dry, metal or glass containers, sealed, weighed, marked, and forwarded to the testing laboratory designated by the activity concerned.

4.1.3 *Sampling for inspection of containers.*—A random sample of filled containers shall be taken by the Government inspector in accordance with Military Standard MIL-STD-105 at Inspection Level I and Acceptable Quality Level = 2.5 percent defective to verify compliance with this specification regarding fill, closure, marking, and other requirements not involving tests.

4.2 Inspection.

4.2.1 *Inspection of containers.*—Each sample filled container shall be examined for defects of construction of the container and the closure.

for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests, provided the contractor has removed or repaired all nonconforming containers.

4.3 **Lot acceptance tests.**—The sample specimens selected in accordance with 4.1.2 shall be subjected to the tests specified in 4.4. Lots shall be accepted or rejected by the Government inspector on the basis of the laboratory test report on the transmitted samples. Rejected lots may be resubmitted for acceptance tests, provided the contractor has removed or reworked all nonconforming products.

4.4 Test procedures.

4.4.1 *Composition.*—The composition of the soap shall be determined according to the methods in Federal Specification P-S-536 as identified in table II.

TABLE II.—*Test methods*

Determination	Method No.
Moisture and matter volatile at 105°C. ..	20.1
Alcohol-insoluble matter	30.1
Free alkali as NaOH	30.2
Free acid as oleic acid	30.2
Matter insoluble in H_2O	30.3
Anhydrous soap	40.1
Sodium chloride	50.1
Rosin	70.1
Titer of mixed fatty acids	90.1
Iodine number (Wijs) of mixed fatty acids	110.1
Sodium silicate as $1Na_2O:3.25SiO_2$	130.1
Sugar	170.5
Residue retained on No. 12 sieve	210.1

4.4.2 *Copper test.*—Copper shall be determined in accordance with tentative method Da 30-53 of the Official and Tentative Methods of the American Oil Chemists' Society.

5. PREPARATION FOR DELIVERY

5.1 Material for immediate use.

5.1.1 *Packaging and packing.*—The soap shall be packaged and packed according to good commercial practice and permitting delivery in a satisfactory usable condition at the lowest tariff rates.

5.1.2 *Marking.*—The shipping container shall be marked as specified in Federal Specification PPP-P-32. In addition, each shipping container shall be marked with the tare weight.

5.2 Material for domestic shipment and storage.

5.2.1 *Packaging and packing.*—The soap shall be packed in multiwall sacks conforming to Federal Specification UU-S-48 or fiber drums conforming to Federal Specification PPP-D-723. Unless otherwise specified in the contract or order, the quantity per container shall be 100 pounds.

5.2.2 *Marking.*—Shipping containers shall be marked as specified in Federal Specification PPP-P-31. In addition, each shipping container shall be marked with the tare weight.

5.3 Material for overseas shipment.

5.3.1 *Packaging and packing.*—The soap packaged as specified in 5.2.1 shall be further packed as specified by the procuring agency in the contract or order.

6. NOTES

6.1 Intended use.—The soap covered by this specification is intended for laundering and general cleaning with soft water, where the presence of alkaline salts is not desirable.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein and should specify whether a bid sample is required. (See 3.2, 3.3, 5.1, 5.2, 5.3, and 6.4.)

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to

determine compliance with this specification. If, for any particular purpose, bid samples are required, invitations for bids should state that each bidder shall submit, in time to arrive prior to the date of opening of bids, a sample of the soap he proposes to furnish, to show form, color, odor, and condition, and should specify the size and weight of sample unit required. Samples should be marked with the name of the commodity, the schedule number and date of opening of bids, and the name and address of the bidder. The accepted bid samples should be kept in airtight containers for later comparison with samples from deliveries.

6.4 **Basis of purchase.**—Powdered soap is subject to a possible gain or loss of weight, depending on atmospheric or storage conditions, or both, or on packaging, as a result of fluctuation in the moisture content. Changes in the moisture content result in a corresponding change in the percentage of total solids or anhydrous soap content, or both.

6.4.1 The material should be purchased by net weight, provided the matter volatile at 105°C. is neither above nor below 5 percent.

6.4.2 Deliveries containing more than 7 percent of matter volatile at 105°C. should be rejected without further test.

6.4.3 On deliveries containing 7 percent or less of matter volatile at 105°C. settlement should be made on the basis of a product containing 5 percent moisture; that is, 0.95 pound of nonvolatile matter should be considered 1 pound of soap. The net weight of the material to be paid for on a 5 percent moisture and volatile matter basis is calculated as follows:

$$W = \frac{R \times (100-L)}{95}$$

where:

W = net weight of material to be paid for on 5-percent moisture and matter volatile basis.

R = net weight of material as received.

L = percentage of loss at 105°C.

95 = percent nonvolatile matter.

For example: For 6 percent moisture and volatile matter where $R = 1$ pound,

$$W = \frac{1 \times (100-6)}{95} = \ldots\ldots \text{ pounds}$$

Cost of soap = number of pounds of soap as received × ...... × price per pound.

Likewise, for soap containing less than 5 percent moisture, such as 4 percent,

$$W = \frac{1 \times (100-4)}{95} = \ldots\ldots \text{ pounds}$$

Cost of soap = number of pounds of soap × ...... × price per pound.

6.5 Transportation description.

Transportation description applicable to this item is:

Form 1

Soap, not otherwise indexed by name, other than liquid

Carload weight 36,000 pounds

Truckload weight, 36,000 pounds.

Form 2

Soap powders

Carload weight 36,000 pounds.

Truckload weight 36,000 pounds.

Minimum weights shown apply when material is transported at classification ratings. They will vary when material moves on commodity rates. When in doubt, consult General Services Administration Regional Traffic Manager.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Copies of this specification may be purchased for 10 cents.